

NEUBURG, Germany

WMO: 108530

Lat: 48.7086N

Lon: 11.2147E

Elev: 396

StdP: 96.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.3	(c) -9.2	(d) -14.0	(e) 1.2	(f) -11.5	(g) -10.9	(h) 1.5	(i) -8.0	(j) 11.8	(k) 6.6	(l) 10.3	(m) 5.5	(n) 1.5	(o) 40	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.4	(c) 30.8	(d) 19.7	(e) 28.8	(f) 19.0	(g) 27.0	(h) 18.3	(i) 20.6	(j) 28.3	(k) 19.7	(l) 26.9	(m) 19.0	(n) 25.6	(o) 2.4	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.6	23.1	17.2	12.9	22.2	16.5	12.3	21.7	61.0	28.4	58.2	26.9	55.5	25.6	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.3	6.2	-15.9	33.6	3.7	1.7	-18.6	34.8	-20.8	35.8	-22.9	36.7	-25.6	37.9
			-16.3	22.1	3.7	0.9	-18.9	22.8	-21.1	23.3	-23.2	23.8	-25.9	24.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-0.5	1.0	5.0	9.3	13.8	17.4	18.9	18.7	14.1	9.1	4.2	0.7
	DBStd	7.88	4.62	4.62	3.94	3.84	3.54	3.56	3.15	3.17	3.29	3.59	3.56	4.27
	HDD10.0	1333	325	253	160	57	8	1	0	0	4	61	176	289
	HDD18.3	3414	583	486	414	270	144	60	32	36	133	286	423	547
	CDD10.0	1097	1	0	3	37	127	222	277	269	126	33	2	0
	CDD18.3	138	0	0	0	0	6	32	50	46	4	0	0	0
	CDH23.3	1581	0	0	0	10	99	364	542	492	72	2	0	0
	CDH26.7	461	0	0	0	1	16	100	176	160	9	0	0	0
Wind	WSAvg	2.4	2.7	2.7	3.0	2.6	2.4	2.2	2.3	2.1	2.1	2.1	2.3	2.5
Precipitation	PrecAvg	776	50	41	52	49	78	89	95	81	64	58	60	58
	PrecMax	1036	133	102	152	143	145	149	170	148	134	195	129	129
	PrecMin	520	2	14	15	10	11	34	26	26	14	12	1	6
	PrecStd	108	29	21	32	28	35	29	35	35	32	37	28	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.4	18.9	24.8	28.6	32.0	33.4	33.6	28.0	22.3	15.9	11.9
		MCWB	8.9	9.1	11.0	14.1	17.8	20.8	20.2	20.0	18.0	15.2	12.3	9.2
	2%	DB	9.9	11.1	16.1	22.1	26.1	29.5	30.8	30.5	25.6	19.2	13.1	9.9
		MCWB	7.5	7.5	9.8	13.0	16.9	19.5	19.5	19.7	17.4	14.5	10.3	7.7
	5%	DB	7.7	9.3	13.7	19.7	23.8	27.4	28.5	28.2	23.2	17.0	11.1	8.1
		MCWB	5.8	6.6	8.7	12.3	15.6	18.5	18.9	18.9	16.6	13.4	9.1	6.4
	10%	DB	5.8	7.2	11.3	17.2	21.4	25.2	26.3	26.1	20.9	14.9	9.4	6.2
		MCWB	4.4	5.0	7.8	10.9	14.3	17.5	18.2	18.1	15.4	12.3	7.8	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	9.5	11.8	15.4	18.9	21.8	21.5	21.6	19.4	16.4	12.2	9.6
		MCDB	11.5	13.2	16.8	22.9	26.5	29.6	30.1	29.3	25.4	20.5	15.0	11.3
	2%	WB	7.4	7.8	10.4	13.7	17.4	20.3	20.3	20.4	18.1	14.9	10.6	8.0
		MCDB	9.5	11.0	14.9	20.5	24.7	27.4	28.2	28.1	24.2	18.4	12.6	9.8
	5%	WB	5.7	6.6	9.2	12.6	16.2	19.2	19.6	19.6	17.1	13.8	9.3	6.4
		MCDB	7.4	9.0	13.0	18.5	22.7	26.0	26.8	26.7	22.1	16.7	10.9	7.8
	10%	WB	4.3	5.2	8.0	11.4	15.0	18.2	18.7	18.6	16.0	12.5	7.9	5.0
		MCDB	5.5	7.1	11.1	16.7	20.4	24.3	25.0	24.9	20.2	14.7	9.2	6.1
Mean Daily Temperature Range	5% DB	MDBR	5.4	7.4	9.2	11.1	11.0	11.3	11.4	11.6	10.9	8.9	5.8	5.1
		MCDBR	7.9	10.8	14.5	16.2	15.6	15.4	15.8	15.9	15.2	12.5	9.4	7.3
		MCWBR	6.0	7.4	8.8	8.5	7.4	7.0	6.3	6.5	7.9	7.9	6.9	5.7
	5% WB	MCDBR	7.6	10.3	13.0	14.8	14.0	14.1	13.8	14.3	13.3	11.6	8.7	7.1
		MCWBR	5.9	7.3	8.2	8.0	6.9	6.6	5.9	6.2	7.1	7.6	6.6	5.6
Clear-Sky Solar Irradiance	taub	0.303	0.326	0.360	0.392	0.401	0.405	0.410	0.404	0.379	0.353	0.325	0.299	
	taud	2.439	2.381	2.328	2.268	2.274	2.294	2.273	2.311	2.358	2.423	2.461	2.451	
	Ebn at Noon	766	824	849	855	861	857	846	834	819	781	729	720	
	Edh at Noon	64	85	105	124	128	127	127	117	102	81	63	56	
All-Sky Solar Radiation	RadAvg	0.95	1.80	2.94	4.30	5.04	5.63	5.36	4.67	3.36	2.04	1.05	0.76	
	RadStd	0.14	0.26	0.37	0.57	0.57	0.46	0.48	0.45	0.41	0.25	0.13	0.09	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.38	+0.81	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.35	N/A	N/A	+62	+23

Nomenclature: See separate page